

A traditional Japanese garden scene featuring a stone path leading through mossy rocks and cherry blossoms. The path is made of dark, wet stones, and the surrounding area is covered in vibrant green moss. In the background, there are traditional Japanese buildings with dark roofs and red lanterns hanging from the eaves. The scene is set against a backdrop of blooming cherry blossoms, creating a serene and picturesque atmosphere.

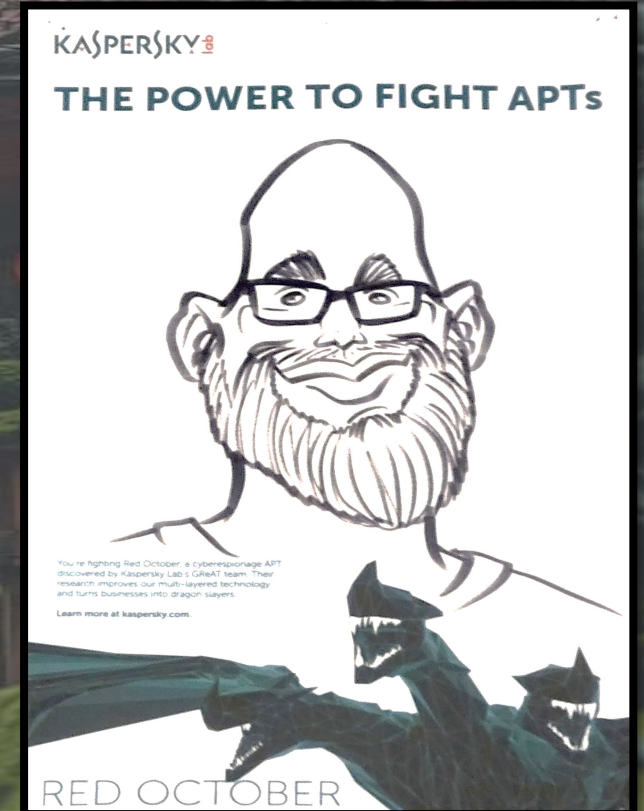
Examining How Geopolitical Conflicts Impact Enterprise Cyber Security

Whoami

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C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 10.0.19043.1889]
(c) Microsoft Corporation. All rights reserved.

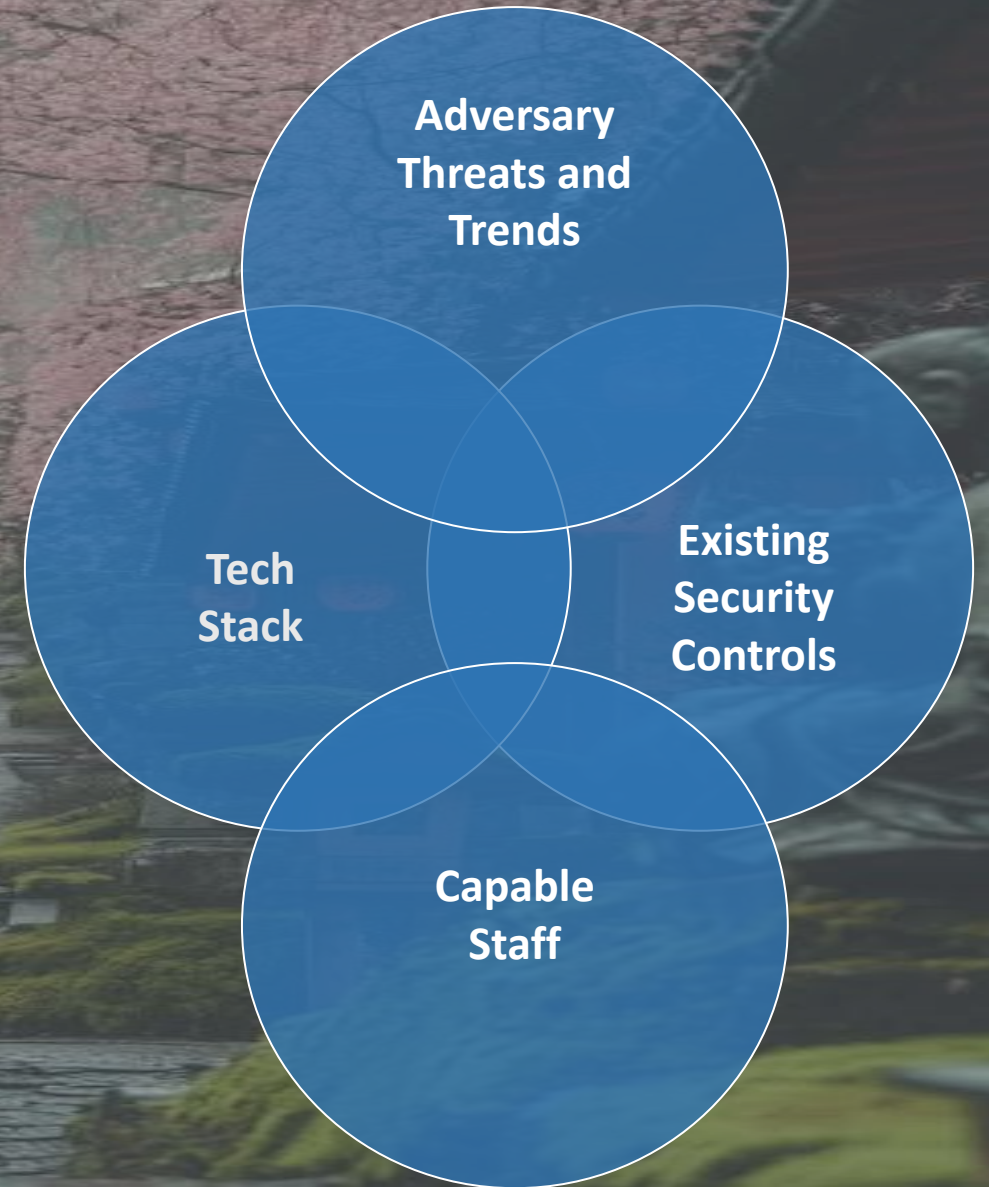
C:\Users\John.Doyle> net user John.Doyle

• SANS - FOR578 Cyber Threat Intelligence Instructor.
• Palo Alto - Director of Cyber Threat Intelligence Services.
• Mandiant - Service Line Lead & Principal Intelligence Consultant.
• Mandiant - Principal Strategic and Incident Response Consultant.
• Mandiant - Principal Advanced Intelligence Integrator.
• Central Intelligence Agency (CIA) - Senior Cyber Threat Analyst.
• George Mason University - Adjunct Cyber Security Professor.
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Organizational Risk and its Drivers

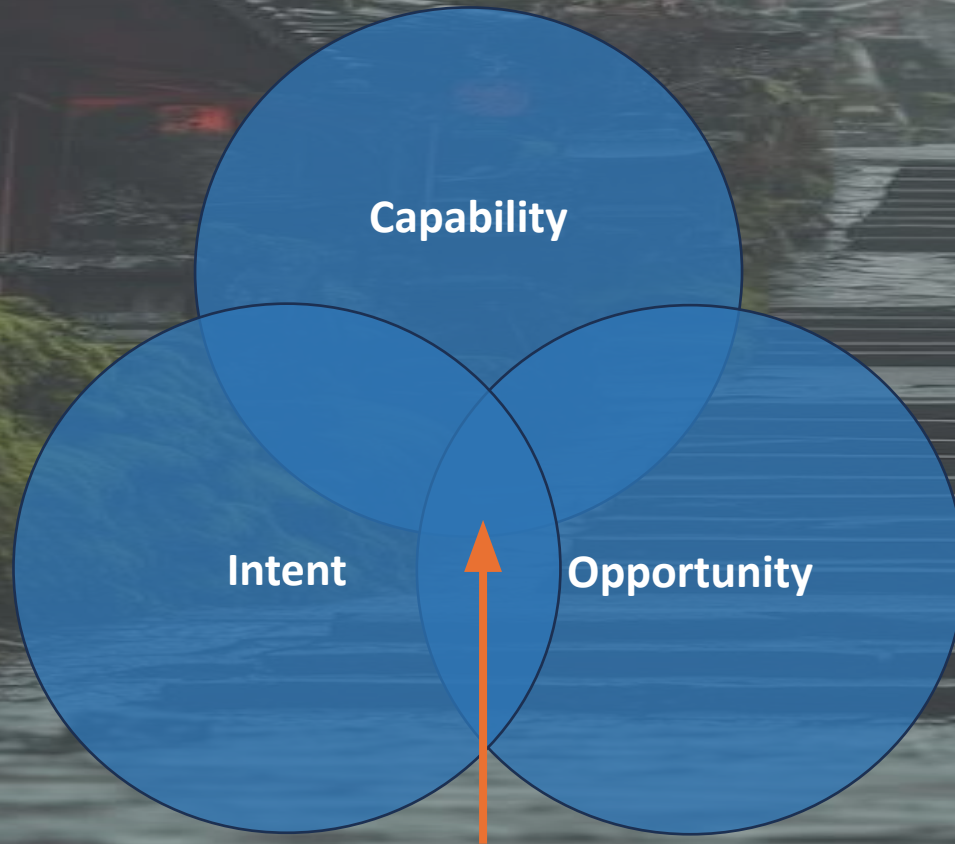
- Threats factor into organizational risk decisions.
- Cyber security teams are often the action arms leverage with intelligence helping provide strategic guidance.
 - Multiple teams usually work on risk exposure reduction and cyber defense
- Organizations can take proactive and reactive steps to reduce risk.
 - Proactive steps reduce risk exposure.
 - Reactive steps reduce accepted risk and account for unexpected unacceptable risks.
 - Ex) Intrusion activity



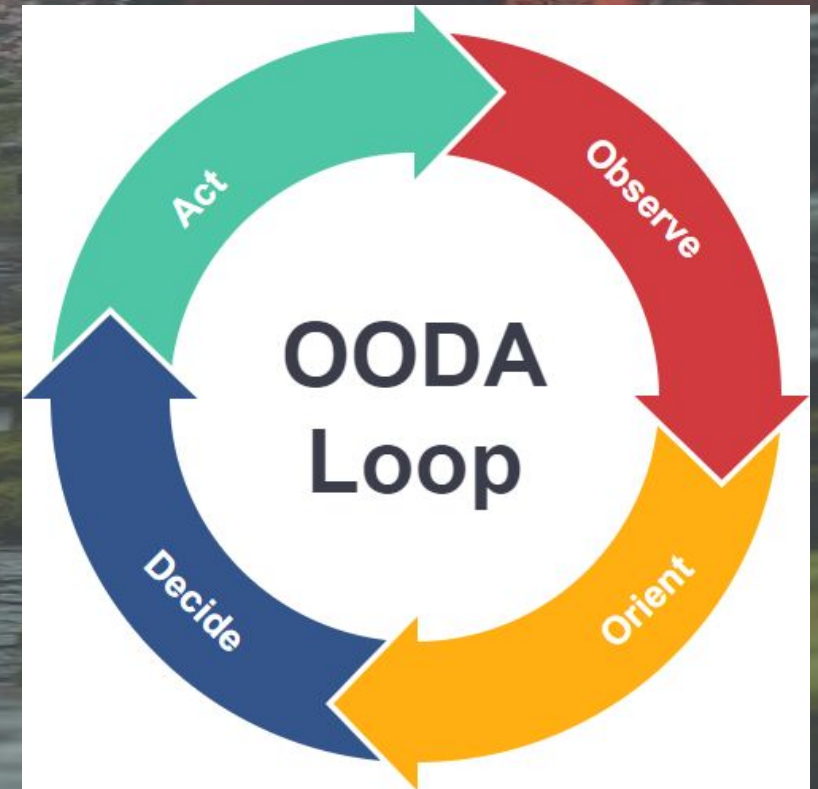
Factors Influencing Risk Exposure

Whack-a-Mole is a Bad Strategy

- Defenders and attackers operate at various speeds to complete objectives. Intelligence speeds up how quickly we can complete the defender's OODA loop



Relevant Threat = Capability + Opportunity + Intent



Cyber as a Tool to Project Power

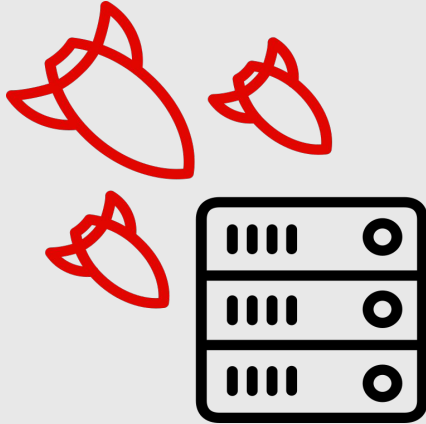
- Geopolitical conflicts often have a cyber component or cyber implications.
- Cyber, as a tool of statecraft, is interwoven across multiple DIMEFIL categories to aid countries in achieving their national-level objectives.

DIMEFIL

Diplomatic		Information	
• Treatises • Bi/multi-lateral Partnerships • International Forums		• Public Diplomacy • Overt and Covert Messaging	
Military		Economic	
• Force Comp • Security • Cooperation and Capacity Building • Show of Force		• Foreign Aid • Trade Agreements • Tariffs • Embargos • Sanctions	
Financial	Intelligence	Law Enforcement	
• Targeted Sanctions • Freeze Assets	• Information Sharing • Partnerships • Collection Operations	• Legal Frameworks • International Partners • Investigation and Prosecution	



Cyber as a Tool to Project Power

				
Gain Insights • Espionage • Counterespionage	Profit Financially • Economic Espionage • Extortion • Fraud • Theft	Shape Behavior • Cyber-Enabled Information Operations • Demonstration of Capability • Exposure • Targeted Attacks	Prepare for War • Operational Preparation	Support War Efforts • Standalone Operations • Force Multiplier Operations

Cyber as a Tool to Project Power

Cyber operations can be used for a variety of purposes:

Espionage

Political

- Provide situational awareness of organizational decisions and plans
- Inform course of action in bi- or multi-lateral negotiations
- Identify potentially compromising information to use as a bartering chip

Military

- Understand capabilities, plans, decision calculus, and key stakeholders

Economic

- Bolster domestic competitiveness
- Fund regime coffers

Attack

- Disrupt or degrade the availability of information systems and network access
- Destroy systems storing information
- In support of concurrent or near-term additional operations
- Cause embarrassment
- Extortion
- Misdirection or as a false flags

...and How it is Often Used



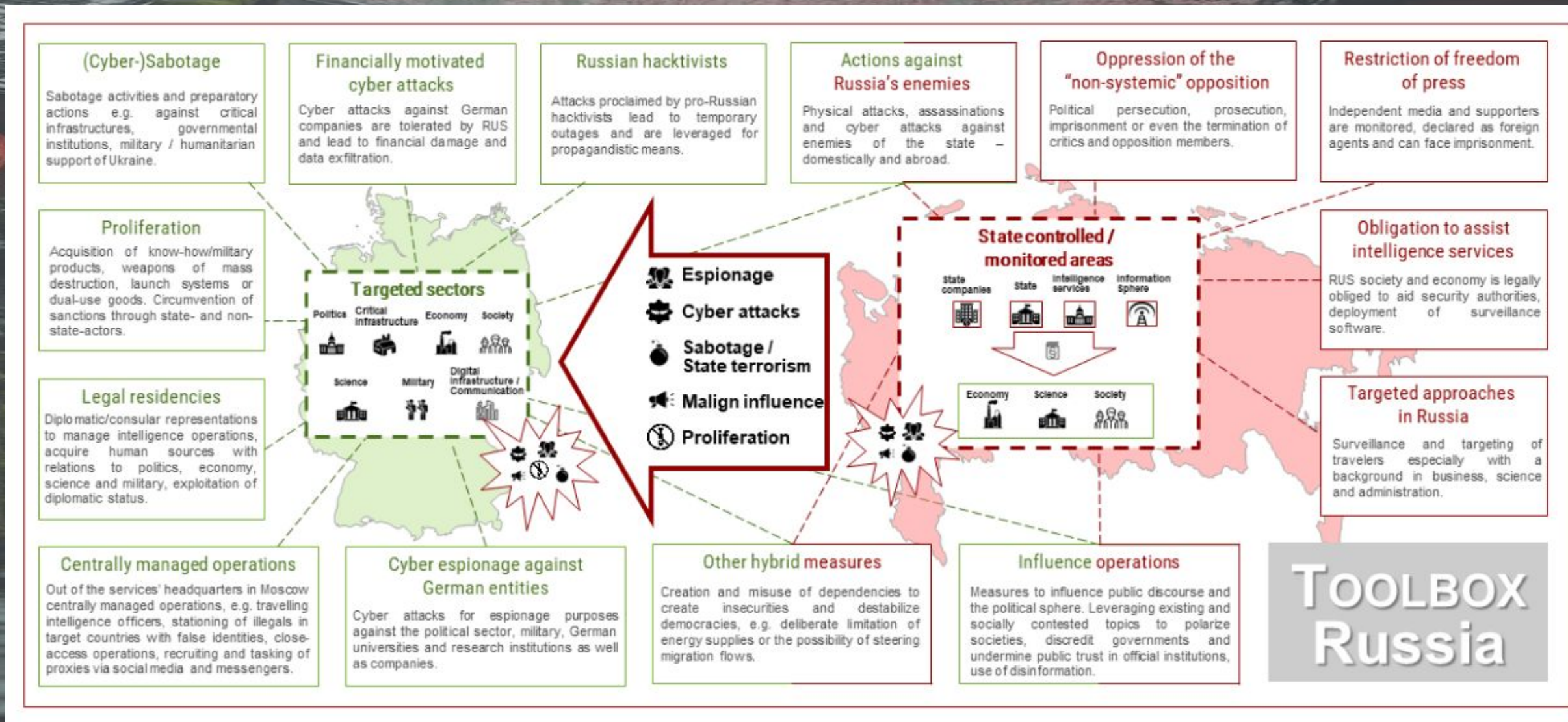
Independent or combined with other war domains operations:

- Espionage
- Disruption (DDoS and wipers)
- Defacement
- Disinformation

Checklist of Questions to Ask Yourself

- Direct impact to business operations?
 - Do we operate in Taiwan or China?
- High risk industry? e.g. Telecommunications, Defense, High Tech, National Gov, Media, or Energy.
- Military or intelligence service contractors or supplier?
- Data transit dependencies?
- Risk level increase for espionage or attack/disruption pre-positioning?
- IO/brand reputation hard to gauge, but keep tabs on public discourse from senior execs commenting on the evolving situation.

...Don't Believe Me, How about the German BfV?



KRIKS | КРИКС
Cryptographic Reconnaissance of
Information and Communication Systems
Криптографическая Разведка
информационно-
Коммуникационных Систем

ITV | ИТВ
Information-Technical
Influence / Effects
Информационно-Техническое
Воздействие

IPV | ИПВ
Information-Psychological
Influence / Effects
Информационно-
Психологические Воздействия

1 Living on the Edge

2 Living off the Land

3 Going for the GPO

4 Disrupt and Deny

5 Telegraphing "Success"

- # SIX PHASES OF RUSSIAN CYBER OPERATIONS DURING THE 2022 WAR IN UKRAINE
- January 2022 - May 2023
-
- The infographic is a horizontal timeline divided into six phases, each represented by a colored circle (blue for Phases I, III, and V; dark blue for Phases II, IV, and VI). Below each phase is a description of the cyber operations. The timeline itself is a horizontal bar with segments for each phase, containing specific operation names and target industry icons. A legend at the bottom right defines the target industries by icon: Government (capitol building), Telecom (phone handset), Financial (stack of coins), Media (megaphone), Energy (lightning bolt), and Transportation (train engine). The timeline segments are: 2019 - JAN 2022 (Phase I), FEB - APR 2022 (Phase II), MAY - JUL 2022 (Phase III), AUG - SEP 2022 (Phase IV), OCT 2022 - JAN 2023 (Phase V), and FEB - MAY 2023 (Phase VI). Operations are listed vertically below the timeline, with lines connecting them to the timeline bar. Some operations are grouped by a bracket. Icons are placed at the end of the operation lines to indicate the target industry.
- | Phase | Description | Timeline Segment | Operations | Target Industries |
|-----------|--|---------------------|--|--|
| PHASE I | Strategic Cyber Espionage and Pre-Positioning | 2019 - JAN 2022 | SHADYLOOK, PAYWIPE | Government |
| PHASE II | Initial Destructive Cyber Operations and Military Invasion | FEB - APR 2022 | INDUSTROYER.V2, CADDYWIPER, SOLOSHRED, AWFULSHRED, JUNKMAIL, CADDYWIPER, PARTYTICKET, SDELETE, NEARTWIST, NEARMISS, PARTYTICKET, SKYFALL | Government, Financial, Media, Energy, Transportation |
| PHASE III | Sustained Targeting and Attacks | MAY - JUL 2022 | CADDYWIPER, CADDYWIPER, CADDYWIPER, CADDYWIPER, CADDYWIPER | Government |
| PHASE IV | Maintaining Footholds for Strategic Advantage | AUG - SEP 2022 | | |
| PHASE V | Renewed Campaign of Disruptive Attacks | OCT 2022 - JAN 2023 | SWIFTSLICER*, CADDYWIPER, ZEROWIPE, SDELETE, AWFULSHRED, BIDSWIPE**, NIKOWIPER*, RANOMBOGGS*, PRESSTEA, DHARMA, CADDYWIPER | Telecom, Financial, Media, Energy, Transportation |
| PHASE VI | Refocus on Strategic Cyber Espionage and Pre-Positioning | FEB - MAY 2023 | WINRAR** | Government |
- Target Industries**
- Government
 - Telecom
 - Financial
 - Media
 - Energy
 - Transportation
- *As reported by ESET
** As reported by CERT-UA

Adversaries are Adaptable

- Strategic doctrine provides a loose outline for enduring collection requirements.
- Shifts in geopolitical relationships (bilateral or multilateral) may change intelligence requirements, shifting cyber resources and targeting priorities.
 - Unexpected actions that take place internationally or domestically may prompt an immediate need to understand on-the-ground realities or prompt retaliation using cyber means.
- Cyber operations calculus – risk tolerance for discovery, attribution, and willingness to be more aggressive – shift from “peace time” to “war time”.
 - Multiple priority requirements likely predicate a shift frequency, volume, and operational targeting. i.e. Expect deviations from previous baselines.
 - There may be a willingness to burn existing access to support immediate mission objectives.
 - New accesses are not instantaneous.
 - Alternative tradecraft and tools may be used, newly emergent threat actor groups created.

		ETERNALPETYA	SOURGRAPES	
BLACKENERGY	INDUSTROYER	BADRABBIT	VPNFILTER	
2015	2016	2017	2018	2019-2021

INDUSTROYER.V2

Wartime Operations (2022 - 2023)

PAYWIPE

SHADYLOOK

PARTYTICKET

NEARMISS

AWFULSHRED

NEARTWIST

SWIFTSLICER

SOLOSHRED

SKYFALL

NIKOWIPER

SDELETE

CADDYWIPER

WINRAR

PRESSTEAL

RANSOMBOGGS

BIDSWIPE

JUNKMAIL

DHARMA

Cyber as a Tool to Project Power

Israel, Iran and Cyber Warfare

MAY 26, 2020, 5:31 PM

Just days after the Israeli media reported earlier this month that Iranian military forces were being withdrawn from Syria, Israelis learned that Iran had sprung a nasty surprise on Israel.

Iran, Israel's most formidable enemy, had launched a cyberattack on Israel's water and sewage facilities. Channel 12 broke the news on May 9, but the actual Iranian hack took place on April 24-25.

Iran's cyber breach of Israel's water command and control systems caused only minor damage, but it set off alarm bells in the Israeli government because Iran had crossed a red line by having targeted a critical component of Israel's civil infrastructure.

...

To no one's surprise, Israel responded forcefully to Iranian aggression. On May 9, Israel launched a cyberattack on the Iranian port of Shahid Rajaei, through which 60 percent of Iran's trade passes to or from the Strait of Hormuz and which houses its most important naval base.

Israel's retaliatory strike, designed to be limited in scope, led to delays in shipments and traffic jams, prompting observers to speculate that the real point of the reprisal was psychological rather than strategic.

Israel's unmistakable message was that its cyber capabilities are among the most advanced in the world and that further Iranian attempts to sabotage its infrastructure would be far more costly to Iran.

Iran gas stations reportedly hit by massive cyberattack

Hackers in Iran address Iran Supreme Leader Ayatollah Ali Khamenei demanding, 'Where is the gas?'

By YONAH JEREMY BOB
OCTOBER 26, 2021 21:41



Gas stations across Iran malfunctioned on Tuesday, reportedly due to a massive cyberattack, according to Iranian state media.

With the details still hazy, speculation is rife about whether the purported attack came from the US, Israel or from local Iranian anti-regime groups.

According to reports, messages were posted in some systems that were [hacked](#), addressing Iran's Supreme Leader Ayatollah Ali Khamenei directly and demanding to know, "Where is the gas?" The attack comes some two years after nationwide protests over gas shortages in fall 2019.

Satellite hack on eve of Ukraine war was a coordinated, multi-pronged assault

The satellite hack that took the world by storm was more complex than initially thought, according to a Viasat executive.

BY CHRISTIAN VASQUEZ AND ELIAS GROLL • AUGUST 10, 2023

LAS VEGAS — The cyberattack that crippled satellite communications on the eve of the Ukraine war was more broad than initially understood and carried out by attackers with detailed knowledge of the compromised system, an executive with Viasat, whose modems were targeted in the attack, revealed during a talk Thursday at the Black Hat cybersecurity conference in Las Vegas.

When hackers attacked Viasat as Russian forces prepared to stream across Ukraine's border, they relied on a piece of malware that wiped the contents of thousands of targeted modems. That component of the attack has been fairly well understood, but on Thursday, Mark Colaluca, vice president and chief information security officer at Viasat Corporate, revealed a second, previously unknown component of the attack and said that the company remains under assault.

That secondary line of attack used "highly technical knowledge of our network" and the networking protocols it relies on to "target specific terminals to not let them back on the network," Colaluca said.

U.S. government and Ukrainian officials have blamed Russia for the attack on the Viasat KA-SAT network that shut down communications that Kyiv relied on for commanding troops and thousands of European used for internet access. And the secondary attack revealed on Thursday showed that once Russian hackers had disrupted internet access in the first place they used sophisticated methods to try and prevent it from being restored.

It's All About the Proxies for Iran

Iranian hostility against Israel date back to at last 1979 with overt tensions starting in the 1990s

- It started when the Supreme Leader disavowed the existence of Israel after taking control of the during the Iranian Revolution

Iran uses proxies to advanced regional objectives through traditional the use of proxies, including against Israel

- Iran provides funding, military aid, advisement, and other support to its regional proxies (Hamas, Hezbollah, Shia, the Houthis, and the Assad regime) aka the “Axis of Resistance”.

Cyber support is independently and in coordination with non-cyber operations.

A “cyber war” has been taking place between Iran and Israel since at least 2020.

Date	Target/Incident	Impact	Perpetrator
24 April 2020	Israel Water Authority systems hacked to increase chlorine levels.	Water pumps at one station malfunctioned, but water supply not contaminated.	Western, Israeli officials attribute to Iran.
9 May 2020	Wiper at Shahid Rajaei Port.	Port operations interrupted for at least three days.	US and other foreign officials attribute to Israel.
20 Nov 2020	Insurer Shirbit database hacked w/ information on Israeli gov employees.	Data encrypted, stolen; hackers threatened to leak data but only did so partially.	BlackShadow hacker group, believed to be Iran.
9 July 2021	National railways come under cyber-attack.	Train service disrupted by fake cancellations and delays.	Unclear, CheckPoint attributes to regime opposition.
13 Oct 2021	Ransomware attack on Hillel Yaffe Medical Centre.	Administrative and medical systems brought down and locked.	BlackShadow hacker group, believed to be Iran.
26 Oct 2021	Petrol stations forced to close after smartcard services blocked.	Long lines formed at 4,300 stations; traffic disrupted. Digital billboards in Tehran and Isfahan were also compromised.	Iran blames Israel and the US for the attack.
30 Oct 2021	Database of Mor Medical Institute breached.	Medical records of 290,000 people posted on Telegram sites.	BlackShadow hacker group, believed to be Iran.
30 Oct 2021	Database of LGBTQ dating site Atraf breached.	Names, photos, and sexual preferences of thousands of users stolen; \$1 million USD ransom sought.	BlackShadow hacker group, believed to be Iran.

The Iranian Cyber M.O.

Iran has a long standing history of using disruptive or destructive cyber operations to respond to actual or perceived aggressions. Post 7 Oct 2023 efforts were no different...

- According to Microsoft, “As the Israel-Hamas war broke out ... Iran surged support to Hamas ... combining targeted hacks with influence operations...”
- ...By late October, nearly all of Iran’s influence and major cyber actors focused on Israel ... in an “all-hands-on-deck” campaign against Israel.
- ...Unlike previous Iranian cyber attacks, all of its destructive cyber operations against Israel in this war were complemented with online influence operations.”



In the Name of God The Merciful

2023-10-30

We (Cyber Av3ngers) have infiltrated to 10 Water treatment stations located in Hedera, Palmachim, Sorek, Ashkelon, Haifa, Chorazim, Kfar Haruv, Taberiye, Eilat and Daniyal.

Once we obtained access to their network, Established to manipulate, wipe and destruct all industrial equipment such as SCADA systems, PLCs, sensors and HMIs.

To the residents of the occupied territories:

Your and your prisoners of war life is worthless to your rulers!

 Cyber Av3ngers

IRGC-Affiliated Cyber Actors Exploit PLCs in Multiple Sectors, Including US Water and Wastewater Systems Facilities

Last Revised: December 18, 2024

Alert Code: AA23-335A

Overview and History of Threat Actor Activity

CyberAv3ngers is an Iranian IRGC-affiliated cyber persona (also known as CyberAveng3rs or Cyber Avengers) that has claimed responsibility for numerous attacks against critical infrastructure organizations primarily in the water and energy sectors in the United States, Israel, and other countries.^{[1][2][3][4][5][6][7][8][9][10]} Since 2020, CyberAv3ngers has claimed responsibility for cyberattacks in Israel, although several of these claimed compromises of critical infrastructure organizations in Israel are false.^[3] In October 2023, CyberAv3ngers claimed credit on their Telegram Channel for cyberattacks against Israel-based PLCs. The PLCs were internet-facing, used Unitronics’ default passwords or no password, and connected to default ports—vulnerabilities that were likely exploited by the actors. CyberAv3ngers also reportedly has connections to another IRGC-linked group known as Soldiers of Solomon. The observed activity includes the following:

- Between Sept. 13, 2023, and Oct. 30, 2023, the CyberAv3ngers Telegram channel displayed both legitimate and false claims of multiple cyberattacks against Israel. CyberAv3ngers targeted Israeli PLCs in the water, energy, shipping, and distribution sectors.
- Beginning on Nov. 22, 2023, IRGC cyber actors accessed multiple U.S.-based WWS facilities that operate HMI-capable Unitronics Vision Series PLCs likely by compromising internet accessible devices with default or no passwords. The targeted PLCs displayed the defacement message, “You have been hacked, down with Israel. Every equipment ‘made in Israel’ is Cyberav3ngers legal target.”

(Update Dec. 18, 2024)

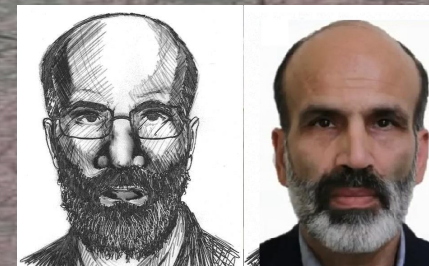
Threat Actor Activity Against U.S.-Based Unitronics Devices

Between November 2023 and January 2024, CyberAv3ngers targeted U.S.-based Unitronics PLC devices used in multiple critical infrastructure industries, including the WWS Sector, likely in four separate waves of cyberattacks. The actors compromised at least 75 devices, including at least 34 in the WWS Sector in the United States.

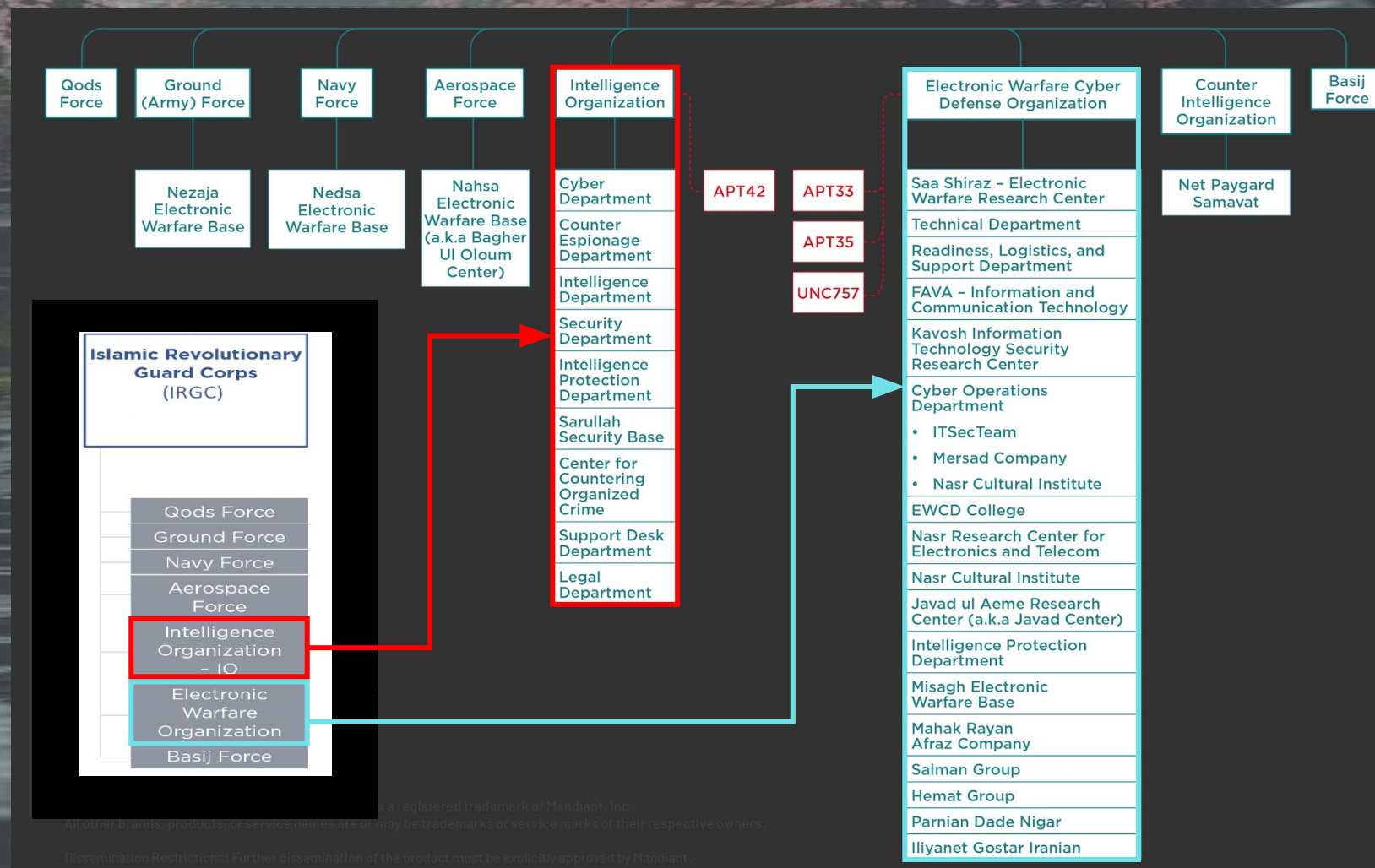
The IRGC's Cyber Program

The IRGC is comprised of the traditional military domain fighting entities – Navy, Ground, Air – as well as electronic warfare and cyber defense (EWCD). EWCD likely controls or influences the cyber-specific R&D efforts for the broader IRGC.

- Electronic warfare units usually are born of signals intelligence (SIGINT) or housed together within a SIGINT unit.
- We assess that the IRGC Electronic Warfare Organization has at least 17 cyber units embedded inside of it.
- IRGC-IO has a unique cyber role in conducting counter espionage, in addition to foreign intelligence collection.



General Lashgarian



China's Historical Context

Taiwan is one of China's "Five Poisons".

- Tensions date back to 1931 with an 18-year civil war between the Chinese Communist Party and the Nationalist Party.
- In December 1949, 2 million members of the Chinese Nationalist Party (Kuomintang) retreated from China to Taiwan. To this day, each claims ownership over the other.
 - The Chinese Nationalist Party called Taiwan the "Republic of China" (ROC), which in response, prompted the Chinese Communist Party (CCP) to call the mainland the "Peoples' Republic of China" (PRC).
- No armistice treaty signed at the end of the Chinese civil war. "Cross-strait relations" is the mutually agreed upon, neutral term by both parties.
- During the 1990s Taiwan elected to advocate to operate as a singular country, moving away from the one-party authoritarian state that had previously committed to eventual unification with China.
 - In response, the PLA attempted to influence the outcome of 1996 Taiwan election, leading to the "3rd Taiwan Strait crisis".
 - The anti-secession law passed by the Chinese National People's Congress in 2005 was a significant escalation point. This essentially reserved the right for China to invade Taiwan should Taiwan attempt to pursue its independence.



Cyber Operations Against Taiwan...

- The Taidoor malware used since at least 2008 is probably the most well-known.
- Second to what took place during and the lead up to surprise visit to Taiwan from U.S. House of Representatives Speaker Nancy Pelosi's amidst rising tensions with China in August 2022...
 - Several Taiwanese government websites were taken offline in a DDoS attack at the rate of 23 times the size of normal traffic loads, according to Taiwan's Digital Minister.
 - The affected websites included the Presidential website, the Ministry of National Defense, the Ministry of Foreign Affairs, and the website of Taiwan Taoyuan International Airport, where Pelosi's plane landed.
 - In addition, 7-11 stores, amongst others, found their TV systems compromised and subsequently defaced, with anti-Pelosi messaging suggesting she leave Taiwan immediately. This took place concurrent to China initiating a military exercise in the region in advance of and during Pelosi's visit to signal displeasure and as a vague threat that it could attack the island at any point in time with its Naval fleet.

Chinese Threat Groups Targeting Taiwan
APT1
APT2
APT5
APT12
APT16
APT17
APT22
APT24
APT41
Conference Crew

...and the Broader Backdrop

- Enactment of laws to potentially amass an arsenal of zero days from Chinese cyber security companies
- Seemingly centralized administration of ORB networks by Chinese companies
- Re-org from the SSF to the ISF
- “Pre-positioning” on critical infrastructure

PRC State-Sponsored Actors Compromise and Maintain Persistent Access to U.S. Critical Infrastructure

Release Date: February 07, 2024

Alert Code: AA24-038A

The Cybersecurity and Infrastructure Security Agency (CISA), National Security Agency (NSA), and Federal Bureau of Investigation (FBI) assess that People's Republic of China (PRC) state-sponsored cyber actors are seeking to pre-position themselves on IT networks for disruptive or destructive cyberattacks against U.S. critical infrastructure in the event of a major crisis or conflict with the United States.

CISA, NSA, FBI and the following partners are releasing this advisory to warn critical infrastructure organizations about this assessment, which is based on observations from the U.S. authoring agencies' incident response activities at critical infrastructure organizations compromised by the PRC state-sponsored cyber group known as Volt Typhoon (also known as Vanguard Panda, BRONZE SILHOUETTE, Dev-0391, UNC3236, Voltzite, and Insidious Taurus):

- U.S. Department of Energy (DOE)
- U.S. Environmental Protection Agency (EPA)
- U.S. Transportation Security Administration (TSA)
- Australian Signals Directorate's (ASD's) Australian Cyber Security Centre (ACSC)
- Canadian Centre for Cyber Security (CCCS), a part of the Communications Security Establishment (CSE)
- United Kingdom National Cyber Security Centre (NCSC-UK)
- New Zealand National Cyber Security Centre (NCSC-NZ)

The U.S. authoring agencies have confirmed that Volt Typhoon has compromised the IT environments of multiple critical infrastructure organizations—primarily in [Communications](#), [Energy](#), [Transportation Systems](#), and [Water and Wastewater Systems](#) Sectors—in the continental and non-continental United States and its territories, including [Guam](#). Volt Typhoon's choice of targets and pattern of behavior is not consistent with traditional cyber espionage or intelligence gathering operations, and the U.S. authoring agencies assess with high confidence that Volt Typhoon actors are pre-positioning themselves on IT networks to enable lateral movement to OT assets to disrupt functions. The U.S. authoring agencies are concerned about the potential for these actors to use their network access for disruptive effects in the event of potential geopolitical tensions and/or military conflicts. CCCS assesses that the direct threat to Canada's critical infrastructure from PRC state-sponsored actors is likely lower than that to U.S. infrastructure, but should U.S. infrastructure be disrupted, Canada would likely be affected as well, due to cross-border integration. ASD's ACSC and NCSC-NZ assess Australian and New Zealand critical infrastructure, respectively, could be vulnerable to similar activity from PRC state-sponsored actors.

Questions?



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